Phase changes

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Learning target: Account for energy transfer during a phase change without a temperature rise.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 5, Day 1.*
Temperature change describes a thermal-state change (°C or K); heat is transferred energy (J). They are not the same quantity.

02. Worked example: explain the model
Rising temperature before melting, flat during phase change; plateau length corresponds to 200 J for this amount.

03. Read and interpret the model
Flat segment is phase change; energy changes intermolecular arrangement rather than increasing temperature.

04. Apply the physics
600 J.

05. Explain your reasoning
Need latent heat mL; zero temperature change does not imply zero energy transfer.

06. Review: connect an earlier idea *Review: Unit 4, Day 14.*
24 N toward the natural length.

07. Exit check
300 J.

08. STRETCH: test the model
100 and 300 J; energy changes the material's phase/internal arrangement.

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01. Warm-up: bring back an earlier idea *Review: Unit 5, Day 1.*
No. Internal energy depends on amount of material and microscopic/material/phase structure as well as temperature.

02. Worked example: explain the model
Rising temperature before melting, flat during phase change; plateau length corresponds to 200 J for this amount.

03. Read and interpret the model
Flat segment is phase change; energy changes intermolecular arrangement rather than increasing temperature.

04. Apply the physics
33,400 J melting; 8400 J warming; 41,800 J total.

05. Explain your reasoning
Need latent heat mL; zero temperature change does not imply zero energy transfer.

06. Review: connect an earlier idea *Review: Unit 4, Day 14.*
F=-kx=+12 N, right.

07. Exit check
L = 45,000/0.20 = 225,000 J/kg. The sample releases energy, so Q = -45,000 J.

08. STRETCH: test the model
Melting 66800 J plus warming 8400 J=75200 J; separate the two processes.